

NIFTY50 Historical Trend & Volatility Analysis

Analysis period: 01 Jan 2010 – 05 Feb 2026

1. Overview

This report analyzes NIFTY50 index data from **01 Jan 2010** to **05 Feb 2026** (4,200 trading days). Over this period, the index moved from **5,200.00** to **21,118.37**, an overall change of **306.12%**. The average India VIX (volatility/fear index) over the period was **26.03**.

2. Yearly Performance

Year	Avg Close	Year Return %	Avg VIX	Trading Days
2010	6,065.07	18.84%	28.39	261
2011	8,902.43	33.85%	30.38	260
2012	13,741.79	111.03%	25.29	261
2013	21,800.55	46.65%	20.43	261
2014	23,836.31	-5.34%	18.07	261
2015	23,394.36	-0.06%	20.00	261
2016	23,904.17	-40.73%	31.78	261
2017	15,354.19	-5.77%	38.12	260
2018	12,485.96	-6.64%	34.32	261
2019	12,393.46	0.59%	19.62	261
2020	12,722.93	-3.14%	16.62	262
2021	13,750.16	18.38%	18.08	261
2022	12,947.97	-19.89%	28.63	260
2023	14,099.19	17.46%	34.36	260
2024	15,920.65	31.53%	31.37	262
2025	20,394.78	11.49%	22.00	261
2026	20,780.55	2.57%	17.55	26

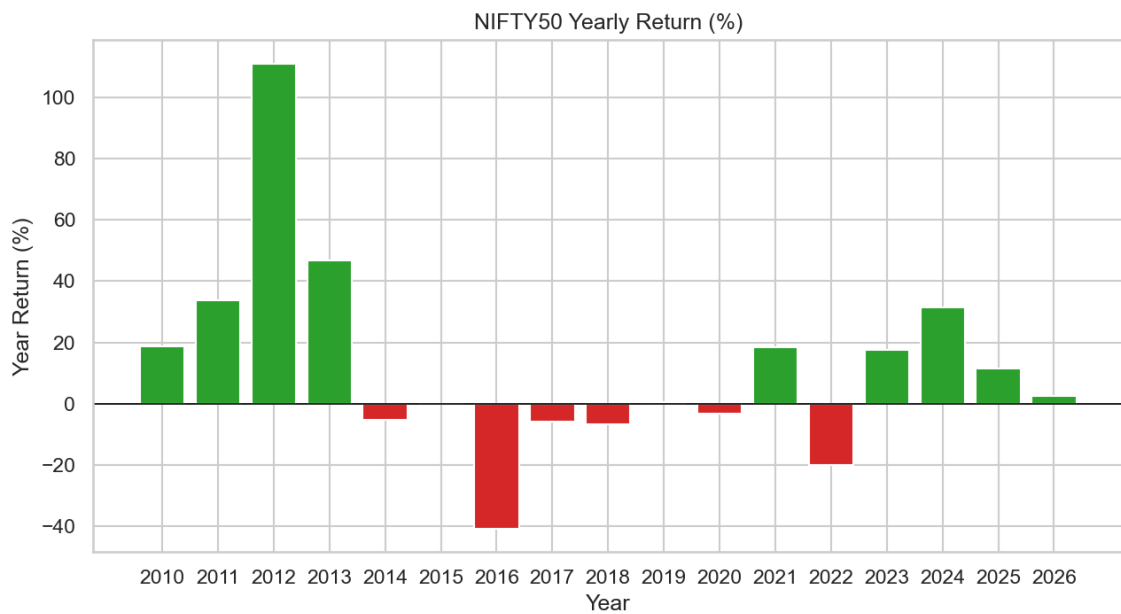


Figure 1: NIFTY50 yearly return (%).

3. Price Trend

The chart below shows the Close price alongside its 50-day and 200-day moving averages, which are commonly used to identify longer-term uptrends and downtrends.

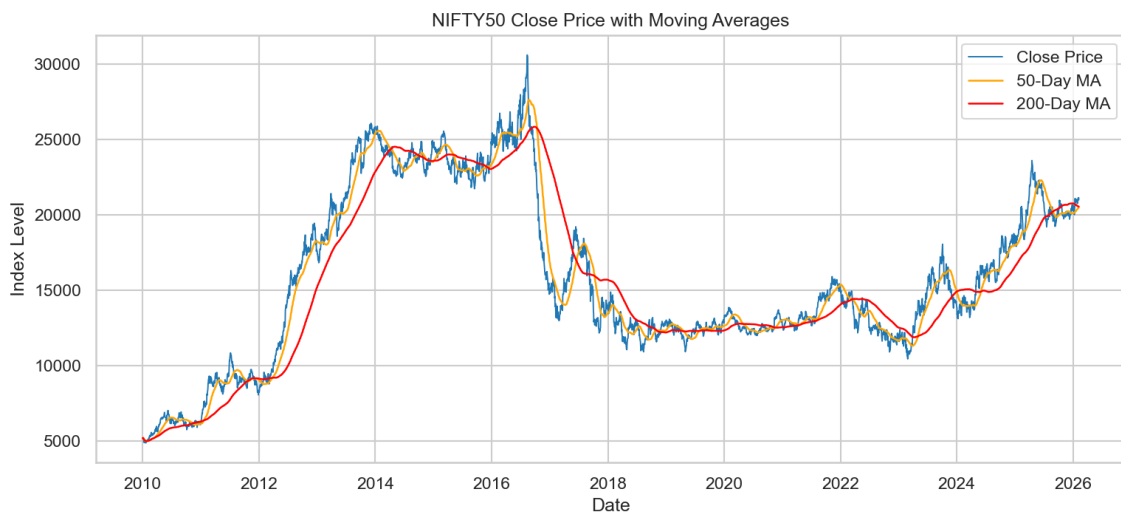


Figure 2: Close price with 50-day and 200-day moving averages.

4. Volatility & Sentiment

Sentiment	Num Days	Avg VIX	Avg Daily Return %
Bearish	2502	30.47	-0.0443%
Bullish	479	17.00	0.3051%
Neutral	1132	20.04	0.1179%
Unknown	87	25.95	0.1653%

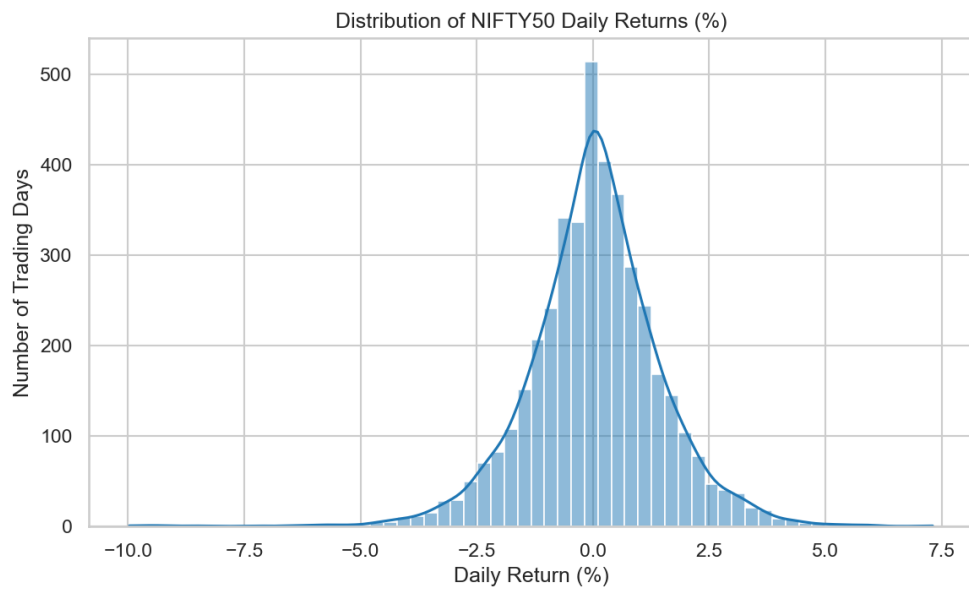


Figure 3: Distribution of daily returns.

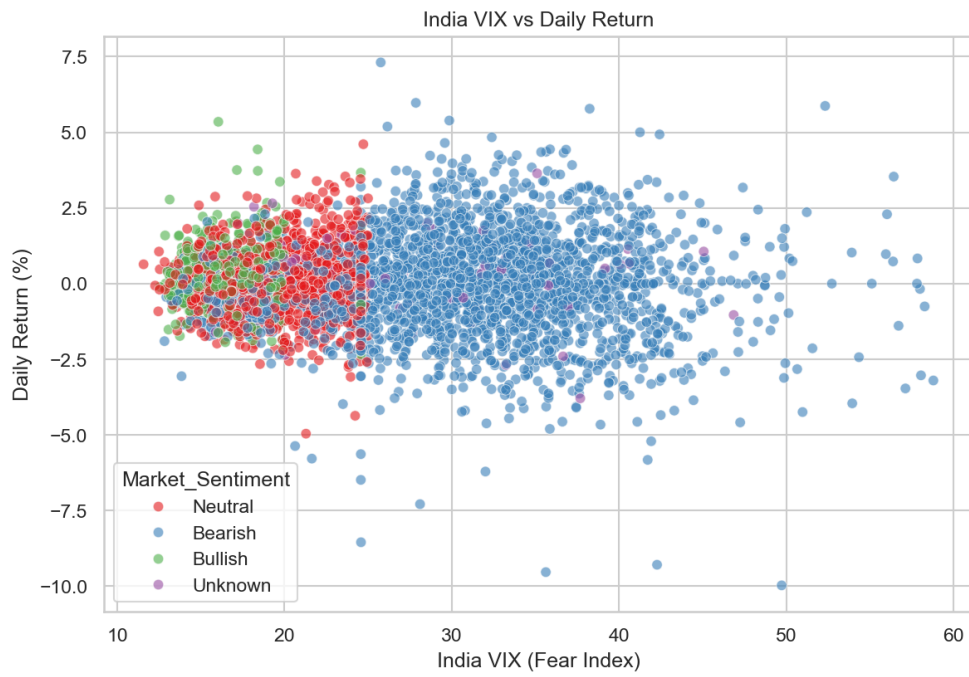


Figure 4: India VIX vs daily return, colored by sentiment.

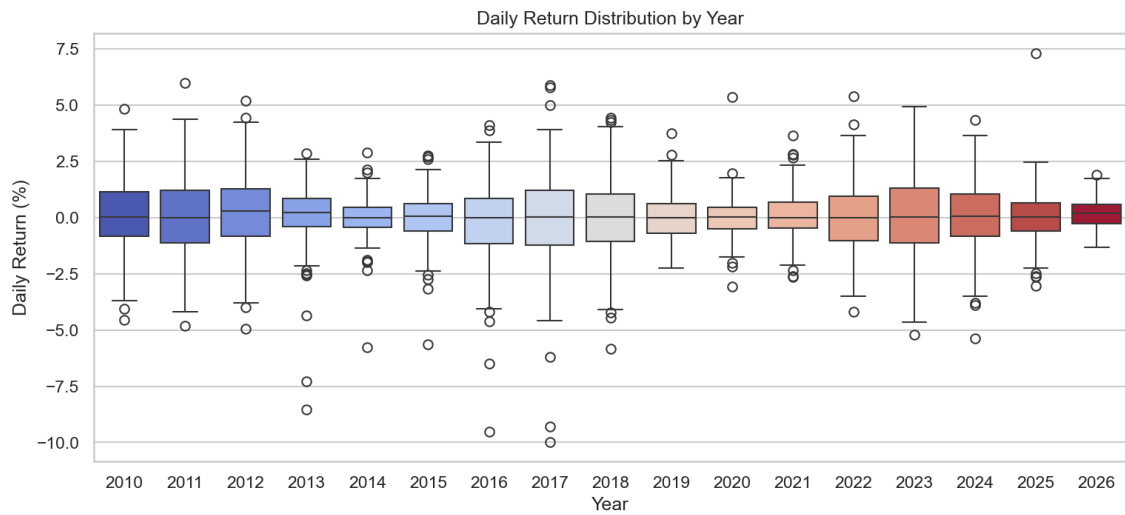


Figure 5: Daily return distribution by year.

5. Sector Trends

Banking was the most frequent top-performing sector over the analysis period.

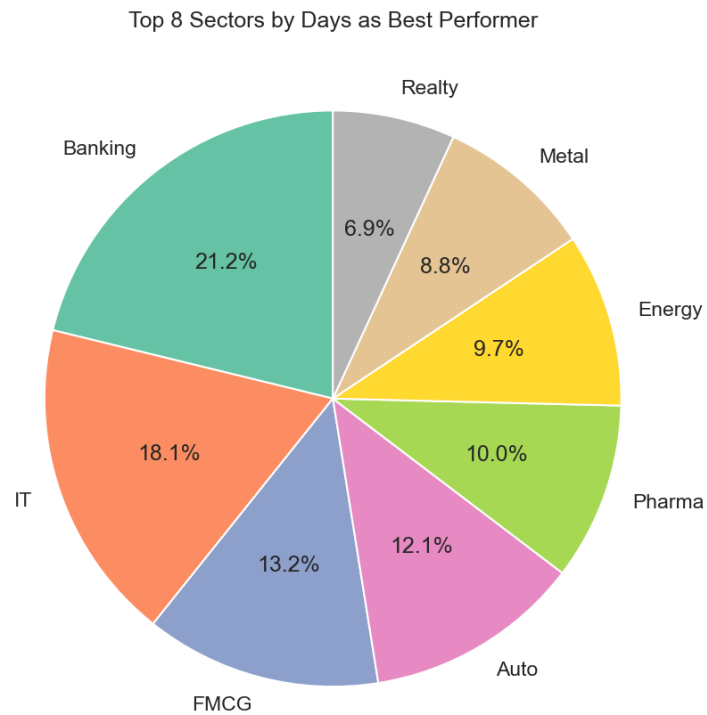


Figure 6: Share of days each sector was the top performer.

6. Relationships Between Indicators

The heatmap and pairplot below show how Close price, Volume, RSI, India VIX and Daily Return relate to one another.



Figure 7: Correlation heatmap between key indicators.

Pairwise Relationships Between Key Indicators

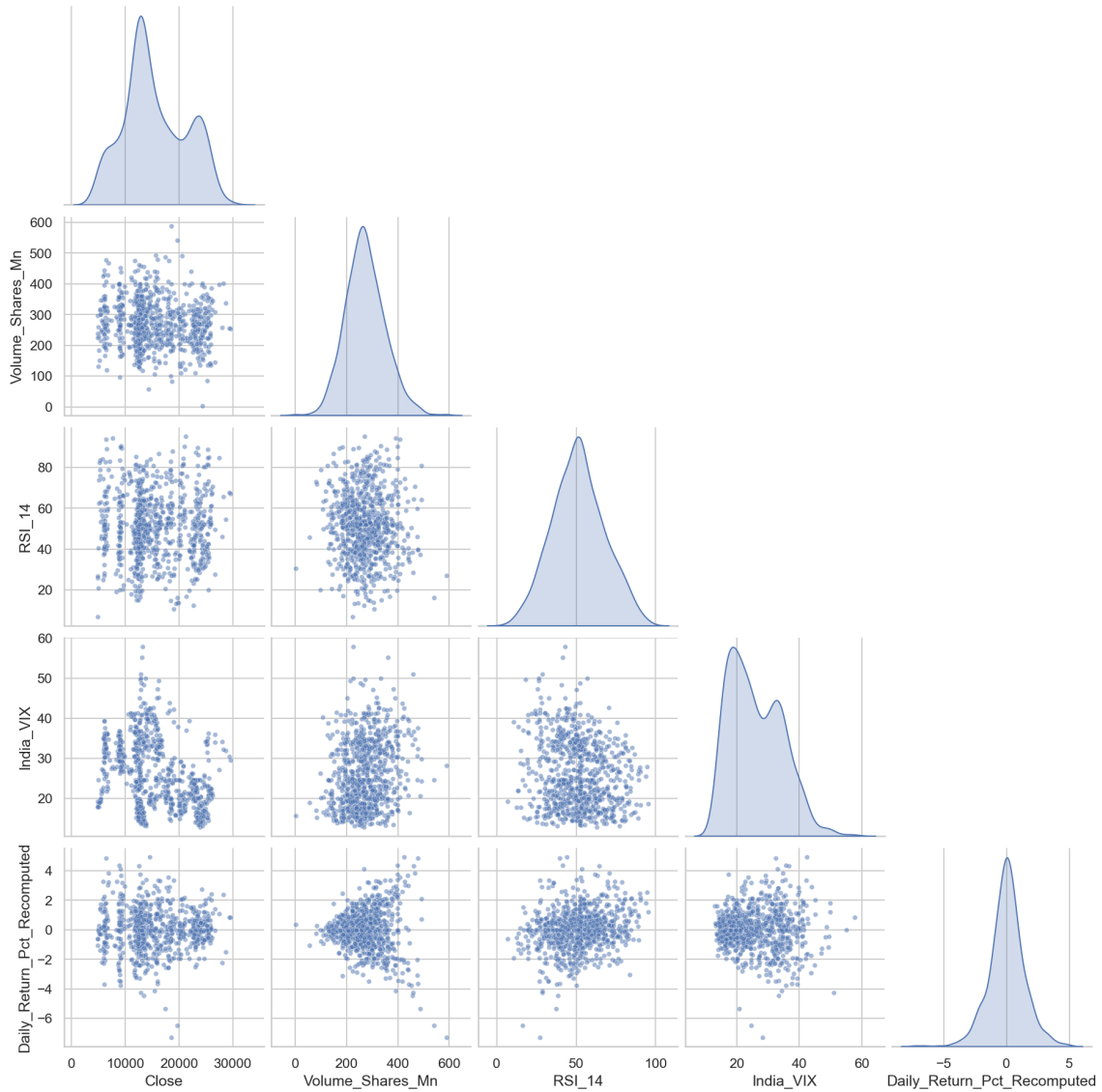


Figure 8: Pairwise relationships between key indicators.

7. Most Volatile Trading Days

Date	Close	Daily Return %	India VIX	Sentiment
02 Oct 2017	13,536.98	-9.97%	49.73	Bearish
01 Jul 2016	25,285.71	-9.53%	35.63	Bearish
13 Sep 2017	14,261.63	-9.29%	42.28	Bearish
03 Oct 2013	22,872.12	-8.55%	24.58	Bearish
06 Feb 2025	20,480.65	7.31%	25.76	Bearish
06 May 2013	18,581.97	-7.29%	28.11	Bearish
21 Oct 2016	19,591.52	-6.49%	24.57	Bearish
23 Aug 2017	15,952.71	-6.21%	32.02	Bearish
08 Feb 2011	7,972.08	5.97%	27.86	Bearish
10 Feb 2017	13,969.82	5.87%	52.33	Bearish

8. Key Takeaways

- Over the full period, NIFTY50 changed by **306.12%**.
- Average volatility (India VIX) stayed around **26.03**.
- **Banking** led as the top sector most often.
- Volatility (return swings) tends to cluster around days with high India VIX, visible in the scatter plot above.
- See `outputs/analysis/business_insights.txt` for the full plain-language summary behind every figure in this report.